

Incident Report: Samba Exploitation Attack (T1210)

Executive Summary

A simulated cyber attack was conducted targeting a vulnerable Samba service on a Metasploitable2 system. The attacker used the Metasploit *usermap_script* exploit to gain unauthorized root access via a reverse shell, resulting in full system compromise. Detection was achieved through correlated behavioral analysis using Wazuh logs and Elastic dashboard visualization. Additional adversary emulation was performed using MITRE Caldera to simulate related attack behavior. The incident highlighted gaps in patch management and monitoring coverage while validating SOC detection and response capabilities.

Timeline

T0: Reconnaissance initiated via network scanning
T1: Samba service exploited using Metasploit
T2: Reverse shell established between attacker and target
T3: Root-level access obtained on target system
T4: Suspicious system activity observed in Wazuh logs
T5: Activity visualized in Elastic dashboard
T6: Incident documented and analyzed in TheHive
T7: Attacker IP blocked using CrowdSec
T8: Adversary emulation executed using Caldera

Root Cause Analysis (RCA)

The primary cause of the incident was an outdated and vulnerable Samba service exposed to the network. The absence of patch management allowed exploitation of a known vulnerability. Additionally, lack of host-based monitoring on the target system limited direct detection, requiring reliance on behavioral correlation from system logs and network activity.

Recommendations

Implement regular patch management and vulnerability scanning across systems. Deploy host-based monitoring agents to improve visibility. Enhance SIEM correlation rules for better detection of exploitation patterns. Restrict unnecessary services and enforce network segmentation. Introduce automated response mechanisms such as SOAR playbooks to reduce response time.

Conclusion

The simulation successfully demonstrated a complete SOC incident lifecycle, including attack execution, detection, analysis, and response. Improvements in monitoring and patch management will significantly strengthen overall security posture.